1071. Greatest Common Divisor of Strings

Easy | Topics | Companies | Hint

For two strings `s` and `t`, we say "`t` divides `s`" if and only if `s = t + t + t + ... + t + t` (i.e., `t` is concatenated with itself one or more times).

Given two strings `str1` and `str2`, return *the largest string* `x` *such that* `x` *divides both* `str1` *and* `str2`.

Example 1:

> **Input:** `str1 = "ABCABC", str2 = "ABC"`
>
> **Output:** `"ABC"`

Example 2:

> **Input:** `str1 = "ABABAB", str2 = "ABAB"`
>
> **Output:** `"AB"`

Example 3:

> **Input:** `str1 = "LEET", str2 = "CODE"`
>
> **Output:** `""`

Example 4:

> **Input:** `str1 = "AAAAAB", str2 = "AAA"`
>
> **Output:** `""`

Constraints:

- `1 <= str1.length, str2.length <= 1000`
- `str1` and `str2` consist of English uppercase letters.

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Leetcode - 1071 - Greatest common / ___
divisor of strings.

We are given two strings:-

str1 = "ABCABC"

str2 = "ABC"

We need to find the largest string that can divide both strings.

↳ but what does dividing a string mean?

↳ in normal mathematics:-

$12 \div 3 = 4$.

for strings, division means that we can repeatedly concatenate a smaller string to construct the bigger string.

for example, "ABC" + "ABC" = "ABCABC"

therefore, "ABC" divides "ABCABC"

"ABC" also divides "ABC".

So, our answer is "ABC".

* We are NOT dividing individual characters, we are finding a common repeating pattern.

Ex

str1 = "ABABAB", str2 = "ABAB"

now, "AB" + "AB" + "AB" = "ABABAB"

"AB" + "AB" = "ABAB"

both strings can be constructed using "AB".
can we use "ABAB"?
No, because "ABABAB" by repeating itself.
therefore, output = "AB".

Ex 3: "LEET" "CODE" → no common repeating pattern, " " → o/p.

Ex 4: "AAAAA", "AAA" → neither has a comparable common repeating pattern
so, o/p = " ".

although both contain A, "AAAAAB" has a B at the end,
Repeating "A" cannot construct "AAAAAB".
hence, the answer is an empty string.

Optimal approach (using GCD + string concatenation)

Obs 1: check whether a common pattern is even possible
consider, str1 = "ABCABC"
str2 = "ABC"
what happens if we concatenate them?
str1 + str2 = "ABCABCABC"] both are
str2 + str1 = "ABCABCABC"] equal.

now consider,
str1 = "LEET"
str2 = "CODE"
concatenate them:-
str1+str2 = "LEETCODE"
str2+str1 = "CODELEET"] different.
this tells us that the strings cannot have a common repeating bar.

If both strings are made using the same repeating block, then, joining them in either order produce the same result.
The common repeating pattern makes the concatenation identical, if they are not identical, a common repeating base is impossible.

Observation 2 ⇒ GCD of string lengths.

consider:- str1 = "ABCABC"
str2 = "ABC"
their lengths are, str1.length() = 6
str2.length() = 3.
find the GCD: GCD (6,3) = 3.
now, take the 1st 3 chars.
str1.substr(0,3) = "ABC"
↓
that's our answer.

why GCD?

if a common string has length k, it must fit perfectly into both strings, therefore, n mod k = 0
m mod k = 0,
so, k must be a common divisor of both lengths.
But we need the largest possible common divisor. that's GCD.
therefore

Answer length = GCD (n, m)

since the common repeating pattern must begin at the beginning of both strings, we simply extract ~~that~~ that many characters from the first string.

```
class Solution { public:
    string gcdOfStrings(string str1,
                        string str2) {
        // comparehelper check.
        if (str1 + str2 != str2 + str1) return "";
        int g = gcd(str1.size(), str2.size());
        return str1.substr(0, g); } };
```

our function accepts two strings & returns a string.

if they match up common,

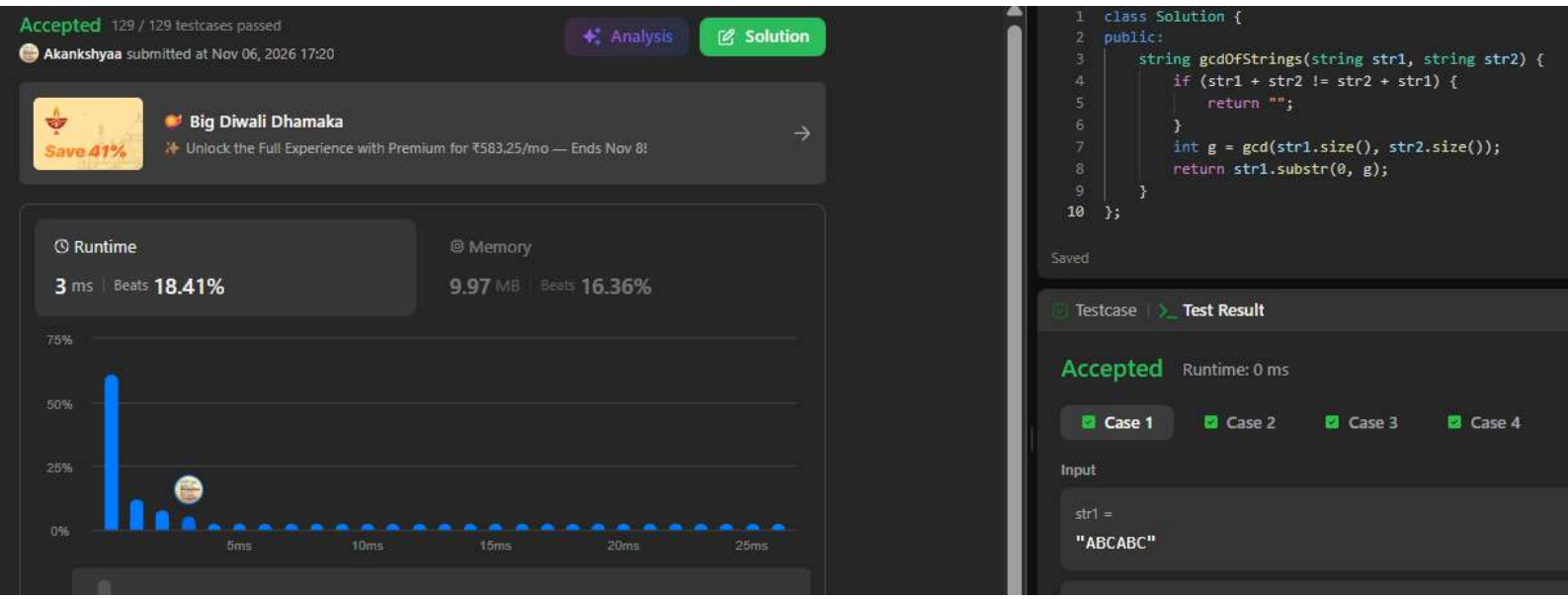
we calculate the gcd of the two given string lengths.

exact answer.

T.C = O(n+m)

↳ str1 + str2 = O(n+m)
↳ str2 + str1 = O(n+m)
↳ gcd(n,m) = O(log(min(n,m)))
↳ substr(0,g) = O(g).

S.C = O(n+m) → concatenation creates temporary strings.
→ the returned substring also requires storage.
→ gcd itself is constant auxiliary space.


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Akankshyaa submitted at Nov 06, 2026 17:20
Analysis
Solution
Save 41%
Big Diwali Dhamaka
Unlock the Full Experience with Premium for ₹583.25/mo — Ends Nov 8!
Runtime
3 ms | Beats 18.41%
Memory
9.97 MB | Beats 16.36%
75%
50%
25%
0%
5ms
10ms
15ms
20ms
25ms
class Solution {
public:
string gcdOfStrings(string str1, string str2) {
if (str1 + str2 != str2 + str1) {
return "";
}
int g = gcd(str1.size(), str2.size());
return str1.substr(0, g);
}
};
Saved
Testcase
Test Result
Accepted
Runtime: 0 ms
Case 1
Case 2
Case 3
Case 4
Input
str1 =
"ABCABC"